MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE TTSU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS

Qualification required.

Construction (Watertight)

First - Copper conductor, uncoated, Navy Standard size 3/5(7).
Second - Extruded silicone rubber insulation. Minimum (min) thickness 0.017-inch.
Third - Insulation jacket of polyamide, type I or type IV of MIL-M-20693, colored (see Fourth). Wall thickness 0.003-inch, nominal (nom). Diameter over insulated and jacketed conductor shall be 0.076-inch, min Twisted Pair Identification code, applied by Method 6.
Fourth - Two conductors (one white, one black), twisted together to form a pair having a lay length not greater than 2-1/4 inches. For TTSU-1-1/2 only, three conductors, (one white, one black, one red), twisted together to form a triad having a lay length not greater than 2-1/4 inches.
Fifth - The specified number of pairs (see Details) cabled together with a lay not greater than 10 inches. Cabling sequence shall be consecutive, starting with pair No. 1, from the center outward. Non-fibrous fillers shall be used to form a firm, well-rounded assembly.
Sixth - Binder or combination binder/barrier.
Seventh - Standard thermoplastic jacket (Gray) (See Details for thickness). Cable surface marking required.

Note 1: When this cable is manufactured with armor, the following additional requirements shall apply:

(a) The type designation shall change to TTSA.
(b) Standard thermoplastic jacket (black or gray) may be used.
(c) Braided metal armor and paint shall be added.
(d) Cable surface marking will not be required.
(e) The maximum overall dimension shall be increased by 0.050-inch.
(f) Group C inspection shall include: Armor - Conformance to material, construction and coverage.

Details

Type and size	Number of pairs	Cable jacket thickness nom (inch)	Overall diameter max (inches)	Cold bending mandrel diameter max (inches)
1/ TTSU-1-1/2	1-1/2	0.050	0.330	5
TTSU-3	3	.050	.450	7
TTSU-5	5	.050	.540	8
TTSU-10	10	.062	.675	9
TTSU-15	15	.062	.800	10
TTSU-20	20	.062	.870	11
TTSU-30	30	.075	1.080	13
TTSU-40	40	.075	1.200	15
TTSU-50	50	.075	1.400	16
TTSU-60	60	.075	1.450	17

1/ TTSU-1-1/2 comprised of three conductors cabled together to form a triad.

(D) Denotes changes

FSC 6145

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(D) IDENTIFICATION CODES:

(a) Twisted pair identification code. This code consists of numbers in sequence running from 1 inch through the number corresponding to the total quantity of twisted pairs in the cable. Both conductors in each pair shall be numbered the same, denoting the sequence number of the pair. Distinction between the two conductors is provided by different colored insulation.

(b) Twisted triad identification code. This code consists of numbers in sequence running from 1 inch through the number corresponding to the total quantity of twisted triads in the cable. All three conductors shall be numbered the same, denoting the sequence number of the triad; distinction between the 3 conductors is provided by different colored insulation.

IDENTIFICATION METHOD:

(D) (a) Method 6. Identification method 6 shall consist of numerals printed in ink on the conductor insulation. For conductors having a jacket directly over the insulation, the numerals may be printed in ink on the jacket, at the manufacturer's option. White ink shall be used for a red or black background; black ink shall be used for a white background. Numerals shall be upright on the longitudinal axis of the conductor (see figure 1). Numeral width shall be proportional to conductor, outside diameter (o.d.) as follows:

Conductor (o.d.) (inch)	Numeral width (inch)
0.040 - 0.050	0.020
.051 - .070	.025
.071 - .095	.030

Numeral height shall be 2-1/2 to 3 times numeral width. Each numeric legend shall be underlined. Two-digit legends shall have the bottom numeral underlined. Legends shall be alternately inverted and shall be repeated at intervals not greater than 1-1/2 inch.

(D)

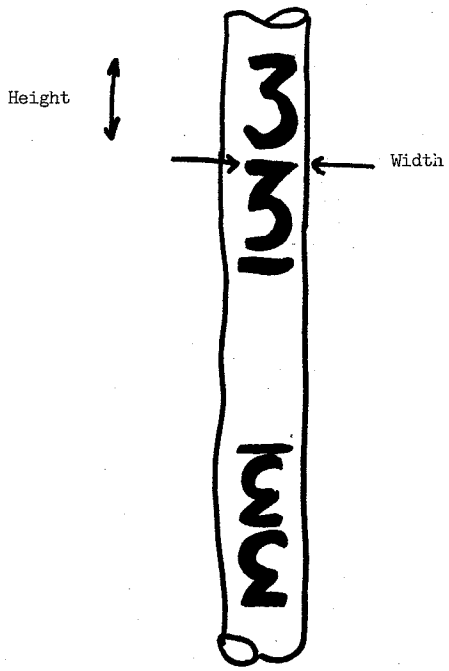


FIGURE 1. Conductor of pair #33.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - At 25° Celsius (C), max ohms/1000 ft.	16.94
Voltage withstand - Volts, root mean square (Vrms), min	
Conductor to conductor.	2000
Conductor to ground	1000
Insulation resistance - Megohms/1000 ft., min.	100
Group A:	
Visual and dimensional	
Watertightness - See MIL-C-915, table of limits for water leakage	
Crack resistance	No damage
Group B:	
Gas flame 1 hour (note 2).	No failure
Drip - At 95°C ± 1°C..	Zero
Cold bending, cable - (See Details for mandrel diameter)	No damage
Physicals (unaged)	
Insulation	
Tensile strength - pound-force per square inch (lbf/in^2), min.	750
Elongation - Percent, min.	125
Jacket	
Tensile strength - lbf/in^2, min.	1800
Elongation - Percent, min.	225
Group C:	
Physicals (aged)	
Jacket	
Hot oil immersion	
Tensile strength - lbf/in^2, min	1400
Elongation - Percent, min	50
Permanence of printing (conductor) - Cycles, min	50
Permanence of printing (jacket), - cycles, min	250
Cable filler removability	

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical, all of Group A, B and C plus the following:

Cable aging and compatibility (125°C ± 5° C.).	No failure
Cold working (minus 20°C ± 2°C).	No damage
Specific gravity of extruded silicone insulation max	1.55

Note 2: Braided armor, when present, shall be connected to neutral terminal of power supply. For gas flame test of TTSU 1-1/2, the three conductors shall be connected to phases A, B, and C of the power supply. For test of all other sizes, connection of conductor pairs shall be in accordance with the following pattern:

Pair #1: One conductor to phase A, one conductor to phase B
Pair #2: One conductor to phase C, one conductor to phase A
Pair #3: One conductor to phase B, one conductor to phase C
Pair #4: One conductor to phase A, one conductor to phase B
Pair #5: One conductor to phase C, one conductor to phase A
Pair #6 through Pair #60: In similar sequence

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UNIT ORDERING LENGTHS

Type and size	Feet (nom)
TTSU-1-1/2 through -10	2500
TTSU-15 and -20	1500
TTSU-30 through -60	1000

Custodians:
Army - MI
Navy - SH
Air Force - 99
Review activities:
Army - EL, AV
Navy - EC
User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0709)

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